Smart Mining System

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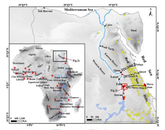
Abstract

As the world progresses and advances in technology development and uses of technology in solving challenges, Egypt must take serious steps to catch thriving countries. Egypt confronts grand challenges in improving scientific and technological base, eradicating public health and facing pollution. Our project mainly is a smart mine and specifically provides solutions for workers' safety and monitoring the pollution caused by emissions from the mining process. Arduino (micro controller) and integrated sensors are monitoring the mine environment. For pollution monitoring MQ-2 is measuring CO2, MQ-135 is measuring methane, DHT22 sensor is used for temperature and humidity monitoring, flame IR for fire detecting, Sharp-GP2Y1014AU0F-Dust-Sensor is used to determine the concentration of dust in the surrounding and ADXL345 is responsible for measuring the change in the three axes X, Y and Z in the case of earthquakes also measuring the change in the pitch in the case of land mass. For data collection ESP32 is used as a Wi-Fi module to send the data collected from the Arduino to the firebase to collect all the sensors reading in a timestamp then by using flutter (with dart backend) all these data are visualized (continuously updated data) in a mobile application. And for data storage SD-card (32GB) is connected using SD-card module in which a csv file is uploaded to this card, and it can be accessible in the form of excel sheet by just opening it on a laptop. Then a test plan was held to ensure the ability of the sensors to detect different surrounding variables. Similar conditions are applied to simulate the mine environment. Thresholds were tested and alerting messages were sent to the mobile phone. All the reading of the sensors are collected in graphs as will be shown in the analysis. Finally, the project meets all the desired design requirements. Requirements are divided into functional requirements in which the different variables are detected, performance requirements as response time and accuracy, usability and interface requirements and power requirements. Our project recorded response time in the range of 1≈3 seconds and for power consumption it needs a battery of 12V, 1088.4 ampere.Hour.

Introduction

Egypt faces a lot of grand challenges (eradicating public health, pollution and increasing the scientific and technological base in Egypt). First, to deal with these problems, we must know the definition of each problem. **First**, eradicating public health in workplaces. Work plays a central role In determining health. It provides wages and benefits, shapes life opportunities and resources for individual workers, their families and communities. **Second**, pollution, Environmental pollution is increasing day by day and imposing severe and irreversible damage to the world. It is of different types: air, water, soil, noise, light, etc. **Third**, technological and environmental bases, the collective panorama of scientific and technological breakthroughs, discoveries, approaches, tools, and infrastructure is referred to as the scientific and technological environment. To solve these problems many prior solutions were held on by many countries like China smart coal project and south Africa Venetia diamond mine which uses IOT and automated haul trucks that save effort also provide the mine safety environment against different hazards. From the disadvantage the high cost. The project is mainly a smart mine provides solutions for mining problems in Egypt (Aswan, South Sainai and the eastern desert). The solution was selected to solve the previous grand challenges as it mainly consists of two monitoring systems. **First**, emissions monitoring system: sensors are used to measure methane (CH4), CO2 and smoke emissions also temperature and humidity are monitored. **Second**, hazards such as earthquakes, land mass and fires are monitored. **Third**, For the real-time database and communication system ESP32 has a Wi-Fi communication system. The Esp32 collect data from Arduino and then sent to the firebase, from firebase the data would be sent to a mobile application dashboard made by flutter (dart). And to be sure of the efficiency of this project there are some design requirements that must be achieved. **First**, functional requirements are to be sure form the functionality of each sensor. **Second**, performance requirements: test response time and accuracy. **Third**, usability and interface: the visualization of the data must be understandable and easy. **Fourth**, power consumption: to calculate the needed power for six months performance. **Fifth**, data storage: needed capacity for saving data for one year must be calculated.



Fig(1) mining areas among Egypt

Fig(2) China coal mining project

Materials

ESP32 WROOM
Used for Wi-Fi, sending to firebase
450 EGP

Flame sensor
To detect flame
35 EGP

MQ-135
To detect Methane
85 EGP

Arduino Uno
To connect sensors
350 EGP

Flutter
For dashboard
Free

Dust Sensor
To detect dust
430 EGP

ADXL345
To detect Acceleration
250 EGP

MQ-2
To detect CO2
85 EGP

SD Card Adaptor
Connect Esp to SD Card
45 EGP

DHT22
Temperature and Humidity detector
100 EGP

Breadboard
Connect wires
80 EGP

Firebase
Connect dashboard with ESP to read the values
Free

Method & Test Plan

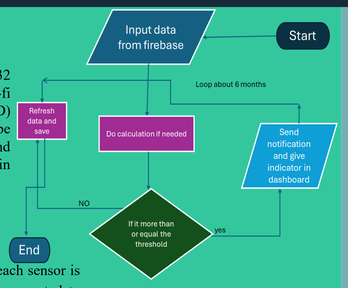
Pre-Requirements:

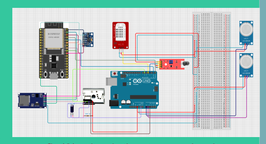
First, Arduino ide must be installed in your device, then esp32 CP210x USB also. **Second**, libraires in Arduino ide (wire – Wi-fi – time - Firebase_ESP_Client- WiFiUdp- NTPClient – SPI – SD) must be installed. **Third**, Visual studio code (VS) must be installed, dart and flutter are installed by downloading file and get installed by terminal. and the extension of flatter and dart in vs code must be added.

For the steps of methods :

Procedures to construct the circuit:

First, Sensors are connected to the Arduino UNO and ESP32 (micro controllers), all are showed in fig.(3) then the code of each sensor is written, and thresholds are tested**. Second** ,Arduino UNO is connected to ESP32 by connecting both TX and RX reversely. Data collected by the Arduino are saved in variables and was sent to the ESP. SD module is connected to the ESP to save data on a csv file that can be opened in the form of an excel sheet when connected to a laptop. **Third** ,ESP was connected to the firebase (firestone) using tokens and URLs. Data is sent continuously and recorded by NTP time format (year-month-day-hour-minute-second). **Forth** , Data was sent to a timestamp in the firestone then this is sent to flutter. **fifth** , User-friendly dashboard was constructed by flutter and the backend was designed by flutter dart. **Sixth** , Data is visualized in the application.





fig(3) stimulation to our circuit

Test Plan Methods :

In our project, the test plan mainly is to test the functionality of each sensor and their ability to detect different factors and alert when the threshold is reached. Test plan is divided into two branches:

1- Testing the ability of sensors to detect the surrounding factors:

I. MQ-2, MQ-135: exposing them to source of methane (lighter) and CO2 (burned paper). Then exposing them to fresh air using a piece of carton. Repeat this and let them return simultaneously.

II. DHT22: exposing it to a hot water vapor to measure both temperature and humidity. Till it reaches a high reading.

III. Sharp-GP2Y1014AU0F-Dust-Sensor: some small sand particles are thrown on the sensitive part of the sensor after covering the rest of it with a tape (repeated).

IV. Flame IR: this was tested by exposing it to the flame of a lighter to output digital values (presence or absence of fire).

V. ADXL345 accelerometer: this was tested by shaking it for a period. Then record values of change in X, Y, Z axes and the pitch.

2- Testing the ability of the mobile application to visualize the data: this was done as follows, while exposing each sensor to its suitable conditions of testing there was a team member monitoring the values appearing on the application and then compare it to the values in the excel sheet.

Results

In our project there are two ways to cast the results: first, using the mobile application and dashboard as shown in the following figures. Second, using an SD card to collect data from the ESP32 in an excel sheet. After the test plan is done, we get results from each sensor to indicate each sensor thresholds, high and low values (as shown in table.1). **For pros**: steady and stable results are sent between ESP, firebase and flutter and maintain as showed in fig 4and a High level of accuracy. **For cons**: first, it was figured out that the Wi-Fi speed has great influence on the speed of sending the data from the firebase to the application. some sensors delay like ADXL345 as shown. Also, humidity is delaying. Second, some sensors gives an outlier value inside the threshold range. Simplified sensor Data were shown in data table 1.

For the scenarios that were conducted, all the system was in a closed box to maintain real live stimulation and reduce noise factors .

Reading	1st	2nd	3rd	4th	5th	6th	7th	8th
Humidity (%)	57.7	60	67	70	89	92	98	100
Temperature (C)	21.6	26	41	50	51.3	41.3	39	38
MQ-135	342	360	535	557	564	403	432	391
MQ-2	374	362	530	580	654	637	645	659
Dust(mg/m3)	0.11	0.29	0.38	0.49	0.29	0.38	0.14	0.14
Change in Pitch (Degree.)	0.22	3.48	9.37	16.93	46.45	82.02	33.04	5.1

Table (1) data of sensors simplified

fig (4) the data appears in application in form of gauge user friendly, and alert is showed in the photo

Analysis

After results were gathered, it was figured out that there is a steady and stable results appear on the application interface also on the excel sheet, some sensors have relatively high accuracy and send data continuous without considerable deflection, application is user-friendly and works appropriately. Gauge changes when any sensor value changes from normal to threshold or higher than threshold and alert message is sent. For negative results, data is sent to the application at a relatively low speed as it depends on the speed of the internet, it was figured also that some data were relatively outliers. Our project aims to solve the problem of environmental issues of Greenhouse gas, and toxicity of heavy metals like sulfur and lead, increasing technological and scientific bases and improving health. Finally, the results met the design requirements that are Power and Communication Requirements, Functional Requirements for sensors reading, Performance Specification and high speed. Usability and User Interface and provide the user historical data.

For sensors accuracy we have used ADXL with 0.25%, DHT22 with 0.2%to 0.5% ,MQ-2 and MQ-135 with 5% ,and dust 0.5V/0.1 mg/m³ (sensitivity to particles).

Earthquake status and Pitch angle:

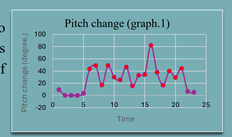
Pitch is rotation about x axis, in our project we use pitch to determine the presence or the absence of land mass and even its strength; to calculate it we use this we calculated the change of the angle around X-axis. This law is to calculate the angle:

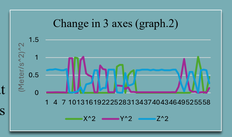
$$\theta = \arctan\left(\frac{-Ax}{\sqrt{(Ay)^2 + (Az)^2}}\right)$$

And here is the law of change:

$$absolute\ Change\ in\ pitch = \left|\frac{\theta 2 - \theta 1}{t2 - t1}\right|$$

If the change in pitch is more than 15 degree this means that there is a possible landmass. The other case, if the change is more than.





45 that mean that land mass is happening now, and workers are in danger. In both cases the decision is left for the owner. The data in the plotted graph (1)have shown the results of the change of pitch that we have carried out

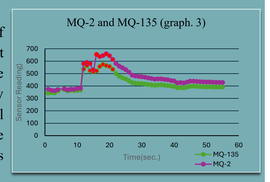
Earthquake calculation:

$$a^2 = (\Delta X)^2 + (\Delta Y)^2 + (\Delta Z)^2$$

In the project we have calculated acceleration by this law to and subtract 0.5 the sensitivity constant, if the "a^2 - 0.5" more than1 and less than 1.35 it will print that there are possible earthquakes and decision maker can act. If it is more than 1.35 decision maker informed that workers are in danger. The graph of X^2 and Y^2 and Z^2 change in the opposite. graph(2)

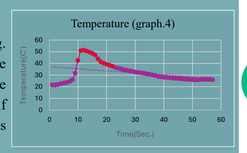
MQ-2 and MQ-135 Gas and Air quality Sensors :

Both sensors have two intervals that determine the level of danger. If MQ-2 reading exceeds 400 the alert will show that level of organic toxins like methane and propane are increasing. If MQ-2 exceeds 600 an alert will trigger to say that workers are in danger. If MQ-135 exceeds 360 it will inform the owner that carbon dioxide is increasing and if the reading exceeds 500 it will show that the environment is getting damaged, and workers are in danger.



Temperature and Humidity :

DHT22 sensor is used for temperature and humidity monitoring. Thresholds were determined; for temperature if the reading of the sensor became more than 35 c° it will send alerting message to the decision maker. Data was represented in graph(4) For humidity, if the humidity increased to 70% alert message is also sent. As showed in graph(5).



Energy and Memory Calculation:

Storage calculation: First the following parameters should be determined, sensor ID, timestamp and measurements value. Sensor ID: 1 byte x 6 sensors = 6 bytes. Timestamp: 4 bytes (shared across all sensors). Measurement value: DHT22: 4 bytes, ADXL345: 6 bytes, Sharp Dust Sensor: 2 bytes, Flame IR: 1 byte, MQ2: 2 bytes, MQ135: 2 bytes. Total 17 bytes. Total storage per reading: 27 bytes. So, the total per year (reading every five seconds):

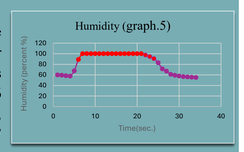
$$27 \times 12(\text{per minute}) \times 60(\text{per hour}) \times 24(\text{per day}) \times 365(\text{per year}) \approx 170.3\ \text{MB}$$

Power calculations: First, total sensors power

$$P = \sum(I \times V) = 2.982\ \text{Watt.Second},$$

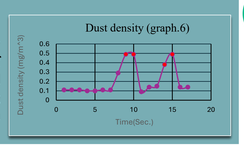
$$\text{Ebattery} = \text{V battery} \times \text{C battery} \quad \text{to calculate it} \quad Runtime(hours) = \frac{E\ battery}{P\ total},$$

$Ebattery = 4380\ hours \times 2.982\ Watt = 13061.16\ W.h$, the battery desired must be 12V, 1088.4 ampere. Hours.



Sharp-GP2Y1014AUOF-Dust-Sensor (pollutants):

Dust sensor's main purpose is to measure the level of heavy metals and dust in the air to inform the owner about the state of the environment. If the reding exceeds 0.35 (mg/m^3) it will inform the decision maker that the environment is getting damaged from heavy metals and dust. All data as showed in graph(6).



Learning Transfer

ST.3.02 We have studied correlation and its types. And how to calculate the residuals and use it to calculate the coefficient of determination (square residuals). As shown in some graphs.

MA.2.07 We learned about derivatives, rates of change, slopes and how to apply these information into a graph. This was helpful in the graph of ADXL345 while calculating the absolute change of pitch.

PH.3.05 We studied communication, Wi-Fi communication, transmitting and receiving signals and other types of wire communication. This helped us in the connection between Arduino, ESP32, firebase and flutter.

ES.3.06 we learned that earthquakes occur due to rapid releasing of stored elastic energy in faults. This energy released in all directions can be detected by the sensor which helps us to avoid this danger by triggering alerts

BI.3.04 we studied spinal cord and the transmission of rapid reflex for immediate response. It has the same function as the smart monitoring system which quickly reflexes to the detection of any hazard by triggering alerts to ensure safety.

Recommendation& Conclusion

Recommendation After finishing the prototype and depending on the analyzed results of the smart mine monitoring system, we have some recommendations for other developers who may work on the same project.1- Improve network connection: as mining operation happens in deep and remote areas, the serial connection of Wi-Fi may be limited or insufficient. So, we recommend using different means of communication such as satellite-based network.2- Using 5G technology instead of 4G: as 5G energy efficient, provides speed up to 10 Gbps and supports more devices. These features are very essential in the process of real-time monitoring.3- Enhance dust sensor: upgrading Sharp-GP2Y1014AU0F-Dust-Sensor to a more accurate sensor as PMS5003.4- Apply fire suppression system: make a connection between the flame IR sensor and automated fire extinguisher system.5- IOT things protection: designing a cover or an enclosure for different electronic components to protect them from harmful particles in the mine environment. 6- We recommend use ML Machine Learning decision tree classifier that make branching in the danger and safety level.

For Conclusion :

By choosing this solution we aimed to help the miners as they exposed to a lot of hazards during mining operations. We addressed this challenge by making a smart mining system which depends on multi-sensors system. After we tested the prototype, we found out that the project met the design requirements and addressed the determined challenges which are workers' safety, real-time data visualization, and an app designed by a simple interface that visualizes data continuously and trigger alerts immediately if the threshold point is reached. Also, it successfully addressed the environmental variables such as toxic gases emissions, heavy metal dust, humidity, and seismic waves activities that refer to the probability of earthquake occurrence. Finally, to get accurate and efficient results we tested the project under stimulated conditions which confirmed the ability of project to detect hazards and trigger real-time alerts within the determined respond time. Also, the requirement of power efficiency has been achieved by the system.

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